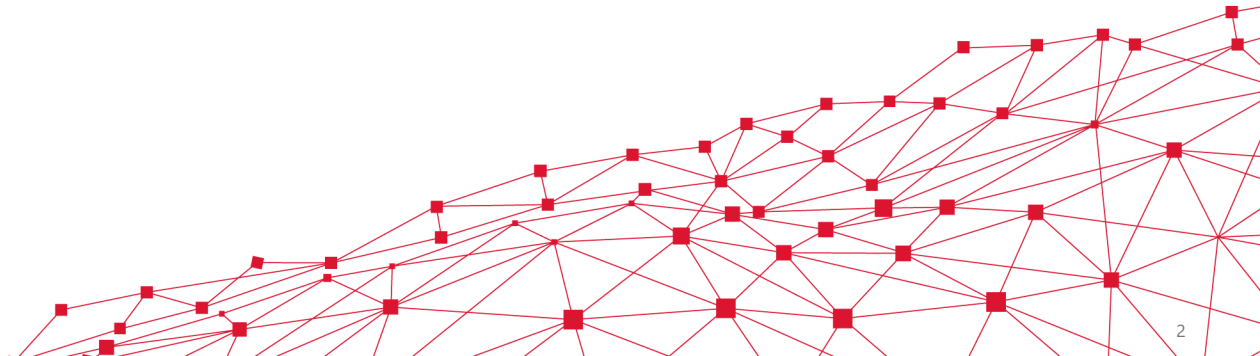




ПОСТРОЕНИЕ ЗАПРОСОВ, ОКОННЫЕ ФУНКЦИИ



ПОДЗАПРОСЫ





ПОДТИПЫ

ОДНОСТРОЧНЫЙ ПОДЗАПРОС

Подзапросы

SELECT DEPTNO, JOB, SAL

← внешний запрос

FROM EMP

WHERE EMPNO =

(SELECT EMPNO FROM EMP WHERE ENAME = 'KING') ;

← подзапрос

1. SELECT EMPNO FROM EMP WHERE ENAME = 'KING' ;

Результат: 7839

2. SELECT DEPTNO, JOB, SAL FROM EMP WHERE EMPNO = 7839 ;

Результат: 10, PRESIDENT, 5000

ОДНОСТРОЧНЫЙ ПОДЗАПРОС



Подзапросы

```
SELECT ENAME  
FROM EMP  
WHERE SAL > (SELECT AVG(SAL) FROM EMP)  
ORDER BY ENAME;  
1. SELECT AVG(SAL) FROM EMP;
```

Результат: 2073,21428571428

```
2. SELECT ENAME FROM EMP WHERE SAL > 2073,21428571428;
```

Результат: BLAKE, CLARK, FORD, JONES, KING, SCOTT

ОДНОСТРОЧНЫЙ ПОДЗАПРОС



Подзапросы

```
SELECT ENAME  
FROM EMP WHERE SAL >  
COALESCE ( (SELECT AVG (SAL) FROM EMP) , 0)  
AND SAL IS NOT NULL -- явно исключаем NULL  
ORDER BY ENAME;
```

```
1. SELECT AVG (SAL) FROM EMP;
```

Результат: 2073,21428571428

```
2. SELECT ENAME FROM EMP WHERE SAL > 2073,21428571428;
```

Результат: BLAKE, CLARK, FORD, JONES, KING, SCOTT

ОДНОСТРОЧНЫЙ ПОДЗАПРОС

Подзапросы

```
SELECT DEPTNO || ' : ' || ROUND(AVG(SAL)) FROM EMP  
GROUP BY DEPTNO HAVING AVG(SAL) <  
(SELECT MAX(AVG(SAL)) FROM EMP GROUP BY DEPTNO);
```

1. SELECT ROUND(AVG(SAL)) FROM EMP GROUP BY DEPTNO;

Результат: 1567; 2917; 2175

2. SELECT MAX(AVG(SAL)) FROM EMP GROUP BY DEPTNO;

Результат: 2917

```
3. SELECT DEPTNO || ' : ' || ROUND(AVG(SAL)) FROM EMP  
GROUP BY DEPTNO HAVING AVG(SAL) < 2917;
```

Результат: 20: 2175
30: 1567



ПОДЗАПРОС В FROM

Подзапросы

```
SELECT DNAME  
FROM (SELECT * FROM DEPT WHERE DEPTNO < 30) ;
```

Результат: ACCOUNTING; RESEARCH

```
SELECT PRDS.PRODUCT_ID, PRICE, PURCHASES_DATA.PRODUCT_COUNT  
FROM PRODUCTS PRDS,  
      (SELECT PRODUCT_ID, COUNT(PRODUCT_ID) PRODUCT_COUNT  
       FROM PURCHASES  
       GROUP BY PRODUCT_ID) as PURCHASES_DATA  
WHERE PRDS.PRODUCT_ID = PURCHASES_DATA.PRODUCT_ID;
```

Результат: 1; 19.95; 4;
 2; 30; 4;
 3; 25.99; 1;

ПОДЗАПРОС В FROM

Подзапросы

```
SELECT ENAME
FROM (SELECT ENAME, SAL, AVG(SAL) as avg_sal
      FROM EMP
      GROUP BY ENAME, SAL) t
WHERE SAL > t.avg_sal ORDER BY ENAME;
```

Результат: BLAKE, CLARK, FORD, JONES, KING, SCOTT;



Подзапросы

Подзапрос пытается передать в оператор равенства (=) во внешнем запросе несколько строк. Пользователь увидит ошибку:

ORA-01427: single-row subquery returns more than one row

В подзапросе указана фраза ORDER BY. Пользователь увидит ошибку:

ORA-00907: missing right parenthesis

МНОГОСТРОЧНЫЙ ПОДЗАПРОС



Подзапросы

```
SELECT ENAME  
FROM EMP  
WHERE EMPNO IN (SELECT EMPNO FROM EMP  
                WHERE ENAME LIKE '%LL%') ;
```

Результат: ALLEN; MILLER

```
SELECT ENAME  
FROM EMP  
WHERE EMPNO NOT IN (SELECT EMPNO FROM EMP  
                   WHERE ENAME LIKE '%A%')  
  
ORDER BY ENAME ;
```

Результат: FORD; JONES; KING; MILLER; SCOTT; SMITH; TURNER

МНОГОСТРОЧНЫЙ ПОДЗАПРОС

Подзапросы

```
SELECT EMPNO, SAL FROM EMP
WHERE SAL > ANY (SELECT SAL FROM EMP
                  WHERE DEPTNO = 10)
AND DEPTNO <> 10;
```

```
SELECT EMPNO, SAL FROM EMP
WHERE SAL > ANY (5000, 2450, 1300)
AND DEPTNO <> 10;
```

```
SELECT EMPNO, SAL FROM EMP
WHERE (SAL > 5000 OR SAL > 2450 OR SAL > 1300)
AND DEPTNO <> 10;
```

EMPNO	SAL
7788	3000
7902	3000
7566	2975
7698	2850
7499	1600
7844	1500

МНОГОСТРОЧНЫЙ ПОДЗАПРОС

Подзапросы

```
SELECT EMPNO, SAL
FROM EMP
WHERE SAL > ALL (SELECT SAL FROM EMP
                  WHERE DEPTNO = 20) ;
```

EMPNO	SAL
7839	5000

```
SELECT EMPNO, SAL
FROM EMP
WHERE SAL > ALL (2000, 3000, 3000, 800, 1100) ;
```

```
SELECT EMPNO, SAL
FROM EMP
WHERE SAL > 2000 AND SAL > 3000 AND SAL > 3000
      AND SAL > 800 AND SAL > 1100 ;
```

МНОГОСТОЛБЦОВЫЙ ПОДЗАПРОС



Подзапросы

```
SELECT DEPTNO, ENAME, JOB, SAL, HIREDATE
FROM EMP
WHERE (DEPTNO, SAL) IN (SELECT DEPTNO, MIN(SAL)
                        FROM EMP GROUP BY DEPTNO)
ORDER BY DEPTNO;
```

```
SELECT DEPTNO, ENAME, JOB, SAL, HIREDATE
FROM EMP
WHERE (DEPTNO, SAL) IN ((10;1300), (20;800), (30;950))
ORDER BY DEPTNO;
```

DEPTNO	ENAME	JOB	SAL	HIREDATE
10	MILLER	CLERK	1300	23.01.1982
20	SMITH	CLERK	800	17.12.1980
30	JAMES	CLERK	950	03.12.1981

КОРРЕЛИРОВАННЫЙ ПОДЗАПРОС



Подзапросы

```
SELECT EMPNO, ENAME, JOB, SAL, DEPTNO
FROM EMP OUTER
WHERE SAL > (SELECT AVG(SAL) FROM EMP INNER
              WHERE INNER.DEPTNO = OUTER.DEPTNO) ;
```

EMPNO	ENAME	JOB	SAL	DEPTNO
7499	ALLEN	SALESMAN	1600	30
7566	JONES	MANAGER	2975	20
7698	BLAKE	MANAGER	2850	30
7788	SCOTT	ANALYST	3000	20
7839	KING	PRESIDENT	5000	10
7902	FORD	ANALYST	3000	20

КОРРЕЛИРОВАННЫЙ ПОДЗАПРОС



Подзапросы

```
SELECT EMPNO, ENAME,  
        JOB, SAL, DEPTNO  
FROM EMP OUTER  
WHERE EXISTS  
    (SELECT EMPNO FROM EMP INNER  
     WHERE INNER.MGR = OUTER.EMPNO) ;
```

```
SELECT EMPNO, ENAME,  
        JOB, SAL, DEPTNO  
FROM EMP OUTER  
WHERE EXISTS  
    (SELECT 1 FROM EMP INNER  
     WHERE INNER.MGR = OUTER.EMPNO) ;
```

EMPNO	ENAME	JOB	SAL	DEPTNO
7902	FORD	ANALYST	3000	20
7698	BLAKE	MANAGER	2850	30
7839	KING	PRESIDENT	5000	10
7566	JONES	MANAGER	2975	20
7788	SCOTT	ANALYST	3000	20
7782	CLARK	MANAGER	2450	10

КОРРЕЛИРОВАННЫЙ ПОДЗАПРОС



Подзапросы

```
SELECT DEPTNO, DNAME  
FROM DEPT  
WHERE NOT EXISTS (SELECT 1 FROM EMP  
                  WHERE EMP.DEPTNO = DEPT.DEPTNO) ;
```

DEPTNO	DNAME
40	OPERATIONS

КОРРЕЛИРОВАННЫЙ ПОДЗАПРОС

Подзапросы

```
SELECT EMPNO, ENAME,  
        JOB, SAL, DEPTNO  
FROM EMP OUTER  
WHERE NOT EXISTS  
    (SELECT EMPNO FROM EMP INNER  
     WHERE INNER.MGR = OUTER.EMPNO) ;
```

```
SELECT EMPNO, ENAME,  
        JOB, SAL, DEPTNO  
FROM EMP  
WHERE EMPNO NOT IN (SELECT MGR FROM EMP) ;
```

EMPNO	ENAME	JOB	SAL	DEPTNO
7876	ADAMS	CLERK	1100	20
7521	WARD	SALESMAN	1250	30
7499	ALLEN	SALESMAN	1600	30
7900	JAMES	CLERK	950	30
7369	SMITH	CLERK	800	20
7934	MILLER	CLERK	1300	10
7654	MARTIN	SALESMAN	1250	30
7844	TURNER	SALESMAN	1500	30

No rows returned.

ВЛОЖЕННЫЕ ПОДЗАПРОСЫ

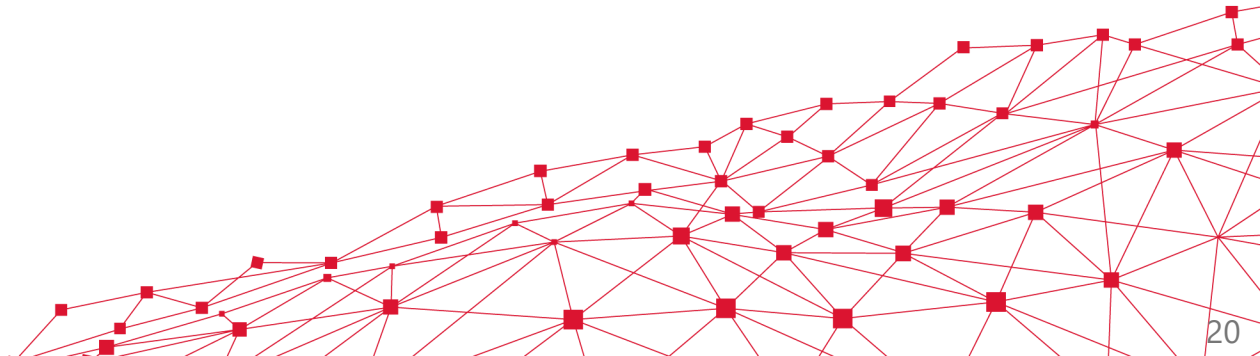


Подзапросы

```
SELECT PRODUCT_TYPE_ID, AVG (PRICE)
FROM PRODUCTS
GROUP BY PRODUCT_TYPE_ID
HAVING AVG (PRICE) < (SELECT MAX (AVG (PRICE) )
                      FROM PRODUCTS
                      WHERE PRODUCT_TYPE_ID IN (SELECT PRODUCT_ID
                                                FROM PURCHASES
                                                WHERE QUANTITY > 1)
                      GROUP BY PRODUCT_TYPE_ID)
ORDER BY PRODUCT_TYPE_ID;
```



ВЫБОР ПЕРВЫХ N-СТРОК



FETCH

Выбор первых N-строк

```
SELECT ENAME, SAL  
FROM EMP  
ORDER BY SAL DESC  
FETCH NEXT 5 ROWS ONLY;
```

ENAME	SAL
KING	5000
SCOTT	3000
FORD	3000
JONES	2975
BLAKE	2850

```
SELECT ENAME, SAL  
FROM EMP  
ORDER BY SAL DESC  
FETCH NEXT 2 ROWS WITH TIES;
```

ENAME	SAL
KING	5000
SCOTT	3000
FORD	3000

FETCH

Выбор первых N-строк

```
SELECT ENAME, SAL  
FROM EMP  
ORDER BY SAL DESC  
OFFSET 5 ROWS  
FETCH NEXT 5 ROWS ONLY;
```

ENAME	SAL
CLARK	2450
ALLEN	1600
TURNER	1500
MILLER	1300
WARD	1250



LIMIT

Выбор первых N-строк

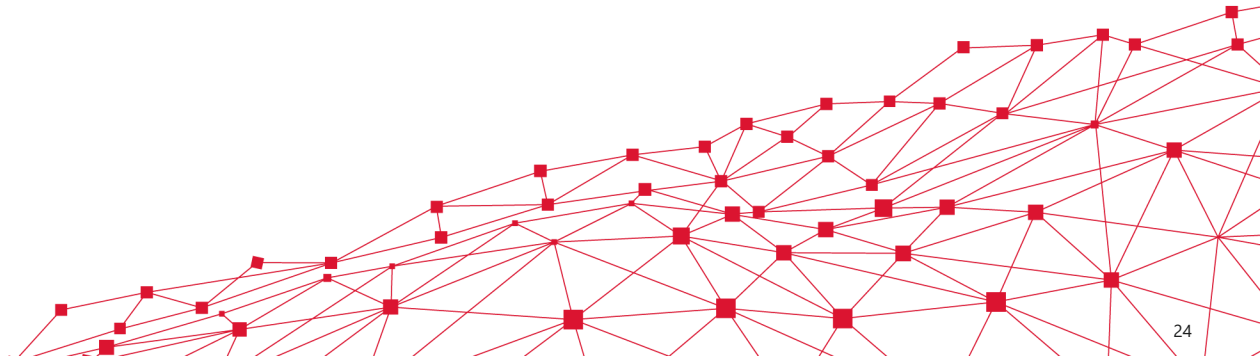
```
SELECT ENAME, SAL  
FROM EMP  
ORDER BY SAL DESC  
LIMIT 5;
```

ENAME	SAL
KING	5000
FORD	3000
SCOTT	3000
JONES	2975
BLAKE	2850





ОКОННЫЕ ФУНКЦИИ



СИНТАКСИС

Оконные функции

```
<window_function>(<table_field>)  
OVER (  
    [PARTITION BY <partition_columns>]  
    [ORDER BY <sort_columns>]  
    [ROWS|RANGE <range_definition>]  
)
```

```
SELECT ENAME, DEPTNO, SAL,  
       ROUND(AVG(SAL) OVER ()) AVG1,  
       ROUND(AVG(SAL) OVER  
             (PARTITION BY DEPTNO)) AVG2,  
       ROUND(AVG(SAL) OVER  
             (ORDER BY DEPTNO)) AVG3  
FROM EMP;
```

ENAME	DEPTNO	SAL	AVG1	AVG2	AVG3
CLARK	10	2450	2073	2917	2917
KING	10	5000	2073	2917	2917
MILLER	10	1300	2073	2917	2917
JONES	20	2975	2073	2175	2453
FORD	20	3000	2073	2175	2453
ADAMS	20	1100	2073	2175	2453
SMITH	20	800	2073	2175	2453
SCOTT	20	3000	2073	2175	2453
WARD	30	1250	2073	1567	2073
TURNER	30	1500	2073	1567	2073
ALLEN	30	1600	2073	1567	2073
JAMES	30	950	2073	1567	2073
BLAKE	30	2850	2073	1567	2073
MARTIN	30	1250	2073	1567	2073

ROW_NUMBER

Оконные функции

```
SELECT DEPTNO, ENAME, EMPNO,
```

```
    ROW_NUMBER() OVER  
      (PARTITION BY DEPTNO  
       ORDER BY EMPNO) EMP_ID
```

```
FROM EMP;
```

DEPTNO	ENAME	EMPNO	EMP_ID
10	CLARK	7782	1
10	KING	7839	2
10	MILLER	7934	3
20	SMITH	7369	1
20	JONES	7566	2
20	SCOTT	7788	3
20	ADAMS	7876	4
20	FORD	7902	5
30	ALLEN	7499	1
30	WARD	7521	2
30	MARTIN	7654	3
30	BLAKE	7698	4
30	TURNER	7844	5
30	JAMES	7900	6



RANK И DENSE_RANK

Оконные функции

```
SELECT DEPTNO, ENAME, SAL,  
       RANK () OVER  
         (PARTITION BY DEPTNO  
          ORDER BY SAL) EMP_ID  
FROM EMP;
```

```
SELECT DEPTNO, ENAME, SAL,  
       DENSE_RANK () OVER  
         (PARTITION BY DEPTNO  
          ORDER BY SAL) EMP_ID  
FROM EMP;
```

DEPTNO	ENAME	SAL	EMP_ID
10	MILLER	1300	1
10	CLARK	2450	2
10	KING	5000	3
20	SMITH	800	1
20	ADAMS	1100	2
20	JONES	2975	3
20	SCOTT	3000	4
20	FORD	3000	4
30	JAMES	950	1
30	MARTIN	1250	2
30	WARD	1250	2
30	TURNER	1500	4
30	ALLEN	1600	5
30	BLAKE	2850	6

DEPTNO	ENAME	SAL	EMP_ID
10	MILLER	1300	1
10	CLARK	2450	2
10	KING	5000	3
20	SMITH	800	1
20	ADAMS	1100	2
20	JONES	2975	3
20	SCOTT	3000	4
20	FORD	3000	4
30	JAMES	950	1
30	MARTIN	1250	2
30	WARD	1250	2
30	TURNER	1500	3
30	ALLEN	1600	4
30	BLAKE	2850	5



Оконные функции

```
SELECT ENAME, HIREDATE, SAL,

      AVG(SAL) OVER
        (ORDER BY HIREDATE ASC
         RANGE 100 PRECEDING) AS
"AVG_SAL_100_DAYS_BEFORE",

      AVG(SAL) OVER
        (ORDER BY HIREDATE DESC
         RANGE 100 PRECEDING) AS
"AVG_SAL_100_DAYS_AFTER"

FROM EMP
ORDER BY HIREDATE DESC;
```

ENAME	HIREDATE	SAL	AVG_SAL_100_DAYS_BEFORE	AVG_SAL_100_DAYS_AFTER
ADAMS	12-JAN-83	1100	2050	1100
SCOTT	09-DEC-82	3000	3000	2050
MILLER	23-JAN-82	1300	2562.5	1300
FORD	03-DEC-81	3000	2340	1750
JAMES	03-DEC-81	950	2340	1750
KING	17-NOV-81	5000	2583.33	2562.5
MARTIN	28-SEP-81	1250	1375	2550
TURNER	08-SEP-81	1500	1975	2340
CLARK	09-JUN-81	2450	2758.33	1975
BLAKE	01-MAY-81	2850	2168.75	2650
JONES	02-APR-81	2975	1941.67	2758.33
WARD	22-FEB-81	1250	1216.67	2358.33
ALLEN	20-FEB-81	1600	1200	2168.75
SMITH	17-DEC-80	800	800	1216.67

Оконные функции

```
SELECT ENAME, HIREDATE, SAL,
```

```
    AVG(SAL) OVER  
      (ORDER BY HIREDATE ASC  
       ROWS 5 PRECEDING) AS  
    "AVG_5_BEFORE",
```

```
    AVG(SAL) OVER  
      (ORDER BY HIREDATE DESC  
       ROWS 5 PRECEDING) AS  
    "AVG_5_AFTER"
```

```
FROM EMP  
ORDER BY HIREDATE;
```

ENAME	HIREDATE	SAL	AVG_5_BEFORE	AVG_5_AFTER
SMITH	17-DEC-80	800	800	1987.5
ALLEN	20-FEB-81	1600	1200	2104.17
WARD	22-FEB-81	1250	1216.67	2045.83
JONES	02-APR-81	2975	1656.25	2670.83
BLAKE	01-MAY-81	2850	1895	2333.33
CLARK	09-JUN-81	2450	1987.5	2358.33
TURNER	08-SEP-81	1500	2104.17	2166.67
MARTIN	28-SEP-81	1250	2045.83	2416.67
KING	17-NOV-81	5000	2670.83	2391.67
JAMES	03-DEC-81	950	2333.33	1870
FORD	03-DEC-81	3000	2358.33	2100
MILLER	23-JAN-82	1300	2166.67	1800
SCOTT	09-DEC-82	3000	2416.67	2050
ADAMS	12-JAN-83	1100	2391.67	1100

Оконные функции

```
SELECT ENAME, HIREDATE, SAL,
-- 5 ДО (включая текущую)
AVG(SAL) OVER (ORDER BY HIREDATE
ASC ROWS BETWEEN 5 PRECEDING AND
CURRENT ROW) AS "AVG_5_BEFORE", --
5 ПОСЛЕ (включая текущую)
AVG(SAL) OVER (ORDER BY HIREDATE
ASC ROWS BETWEEN CURRENT ROW AND
5 FOLLOWING) AS "AVG_5_AFTER"
FROM EMP ORDER BY HIREDATE;
```

ENAME	HIREDATE	SAL	AVG_5_BEFORE	AVG_5_AFTER
SMITH	17-DEC-80	800	800	1987.5
ALLEN	20-FEB-81	1600	1200	2104.17
WARD	22-FEB-81	1250	1216.67	2045.83
JONES	02-APR-81	2975	1656.25	2670.83
BLAKE	01-MAY-81	2850	1895	2333.33
CLARK	09-JUN-81	2450	1987.5	2358.33
TURNER	08-SEP-81	1500	2104.17	2166.67
MARTIN	28-SEP-81	1250	2045.83	2416.67
KING	17-NOV-81	5000	2670.83	2391.67
JAMES	03-DEC-81	950	2333.33	1870
FORD	03-DEC-81	3000	2358.33	2100
MILLER	23-JAN-82	1300	2166.67	1800
SCOTT	09-DEC-82	3000	2416.67	2050
ADAMS	12-JAN-83	1100	2391.67	1100

SUM

Оконные функции

```
SELECT ENAME, DEPTNO, SAL,  
  
       SUM(SAL) OVER  
         (PARTITION BY DEPTNO  
          ORDER BY ENAME) AS "SUM"  
  
FROM EMP;
```

ENAME	DEPTNO	SAL	SUM
CLARK	10	2450	2450
KING	10	5000	7450
MILLER	10	1300	8750
ADAMS	20	1100	1100
FORD	20	3000	4100
JONES	20	2975	7075
SCOTT	20	3000	1007
SMITH	20	800	1087
ALLEN	30	1600	1600
BLAKE	30	2850	4450
JAMES	30	950	5400
MARTIN	30	1250	6650
TURNER	30	1500	8150
WARD	30	1250	9400



FIRST_VALUE И LAST_VALUE

Оконные функции

```
SELECT ENAME, HIREDATE,  
       FIRST_VALUE(ENAME) OVER  
         (ORDER BY HIREDATE ASC  
          RANGE BETWEEN 100 PRECEDING  
                   AND 100 FOLLOWING) AS  
"FIRST_VALUE",  
  
       LAST_VALUE(ENAME) OVER  
         (ORDER BY HIREDATE ASC  
          RANGE BETWEEN 100 PRECEDING  
                   AND 100 FOLLOWING) AS  
"LAST_VALUE"  
  
FROM EMP  
ORDER BY HIREDATE ASC;
```

ENAME	HIREDATE	FIRST_VALUE	LAST_VALUE
SMITH	17-DEC-80	SMITH	WARD
ALLEN	20-FEB-81	SMITH	BLAKE
WARD	22-FEB-81	SMITH	BLAKE
JONES	02-APR-81	ALLEN	CLARK
BLAKE	01-MAY-81	ALLEN	CLARK
CLARK	09-JUN-81	JONES	TURNER
TURNER	08-SEP-81	CLARK	JAMES
MARTIN	28-SEP-81	TURNER	JAMES
KING	17-NOV-81	TURNER	MILLER
FORD	03-DEC-81	TURNER	MILLER
JAMES	03-DEC-81	TURNER	MILLER
MILLER	23-JAN-82	KING	MILLER
SCOTT	09-DEC-82	SCOTT	ADAMS
ADAMS	12-JAN-83	SCOTT	ADAMS

LAG И LEAD

Оконные функции

```
SELECT ENAME, DEPTNO, HIREDATE,  
  
       LAG(HIREDATE, 1) OVER  
         (PARTITION BY DEPTNO  
          ORDER BY HIREDATE) PREVEMPL,  
  
       LEAD(HIREDATE, 1) OVER  
         (PARTITION BY DEPTNO  
          ORDER BY HIREDATE) NEXTEMP  
  
FROM EMP  
WHERE DEPTNO = 20;
```

ENAME	DEPTNO	HIREDATE	PREVEMPL	NEXTEMP
SMITH	20	17-DEC-80	-	02-APR-81
JONES	20	02-APR-81	17-DEC-80	03-DEC-81
FORD	20	03-DEC-81	02-APR-81	09-DEC-82
SCOTT	20	09-DEC-82	03-DEC-81	12-JAN-83
ADAMS	20	12-JAN-83	09-DEC-82	-

Оконные функции

```
SELECT sum(b) OVER w, -- Используем окно 'w'  
       max(b) OVER w -- Переиспользуем то же окно  
FROM t WINDOW w AS (PARTITION BY a ORDER BY b);
```

ФИЛЬТР, ИСКЛЮЧЕНИЕ



Оконные функции

```
SELECT sum(b) FILTER (WHERE b > 1) OVER w, a, b
```

```
FROM t WINDOW w AS (PARTITION BY a ORDER BY b) ;
```

```
SELECT sum(b) OVER (PARTITION BY a ORDER BY b RANGE  
BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW EXCLUDE  
GROUP) , a, b
```

```
FROM t;
```

ФИЛЬТР



Оконные функции

```
SELECT sum(b) FILTER (WHERE b > 1) OVER w, a, b  
FROM t WINDOW w AS (PARTITION BY a ORDER BY b);
```

```
SELECT DEPTNO, ENAME, JOB, SAL,
```

```
--Средняя зарплата всех сотрудников в отделе
```

```
AVG(SAL) OVER (PARTITION BY DEPTNO) AS avg_all,
```

```
--Средняя зарплата только MANAGER'ов в отделе
```

```
AVG(SAL) FILTER (WHERE JOB = 'MANAGER') OVER (PARTITION  
BY DEPTNO) AS avg_managers
```

```
FROM EMP ORDER BY DEPTNO, SAL DESC;
```

DEPTNO	ENAME	JOB	SAL	avg_all	avg_managers	avg_analysts
20	SCOTT	ANALYST	3000	2443.75	2975.0	3000.0
20	FORD	ANALYST	3000	2443.75	2975.0	3000.0
20	JONES	MANAGER	2975	2443.75	2975.0	3000.0
20	SMITH	CLERK	800	2443.75	2975.0	3000.0



Оконные функции

```
SELECT ENAME, HIREDATE, SAL,  
-- Текущая зарплата  
SAL AS current_sal,  
-- Средняя зарплата за 2 предыдущих месяца (только Salesman'ов)  
AVG(SAL) FILTER (WHERE JOB = 'SALESMAN') OVER ( ORDER BY  
HIREDATE ROWS BETWEEN 2 PRECEDING AND CURRENT ROW ) AS  
avg_last_2_salesmen  
FROM EMP  
WHERE JOB IN ('SALESMAN', 'MANAGER')  
ORDER BY HIREDATE;
```



Оконные функции

```
SELECT DEPTNO, ENAME, SAL,  
-- Сколько в отделе получают более 2000  
COUNT(*) FILTER (WHERE SAL > 2000) OVER (PARTITION BY DEPTNO)  
AS high_paid_count,  
-- Сколько в отделе получают менее 1500  
COUNT(*) FILTER (WHERE SAL < 1500) OVER (PARTITION BY DEPTNO)  
AS low_paid_count  
FROM EMP  
ORDER BY DEPTNO;
```

Оконные функции

```
SELECT DEPTNO, ENAME, SAL,  
-- Общая сумма по отделу (все строки)  
SUM(SAL) OVER (PARTITION BY DEPTNO) AS total_dept,  
-- Сумма по отделу БЕЗ текущего сотрудника  
SUM(SAL) OVER (PARTITION BY DEPTNO EXCLUDE CURRENT ROW) AS  
total_without_me  
FROM EMP ORDER BY DEPTNO, SAL DESC;
```

DEPTNO	ENAME	SAL	total_dept	total_without_me
10	KING	5000	10300	5300 (2850+2450)
10	BLAKE	2850	10300	7450 (5000+2450)
10	CLARK	2450	10300	7850 (5000+2850)

Оконные функции

```
SELECT DEPTNO, JOB, SAL,  
-- Стандартная сумма по должности  
SUM(SAL) OVER (PARTITION BY DEPTNO, JOB) AS total_by_job,  
-- Сумма по должности, исключая группы с текущей зарплатой  
SUM(SAL) OVER (PARTITION BY DEPTNO, JOB EXCLUDE GROUP) AS  
total_excluding_my_sal_group  
FROM EMP WHERE DEPTNO = 20 ORDER BY JOB, SAL;
```

DEPTNO	JOB	SAL	total_by_job	total_excluding_my_sal_group
20	ANALYST	3000	6000	0 (исключил всех ANALYST'ов)
20	ANALYST	3000	6000	0 (исключил всех ANALYST'ов)
20	MANAGER	2975	2975	0 (только один, исключил)
20	CLERK	800	800	0